

Vishal Kumar Allam

Dallas, TX | 816-666-2813 | itsmevishal36@gmail.com | **Work Authorization: H1B**

PROFESSIONAL SUMMARY

- Performance-focused **Software Quality Engineer** with **6+ years of experience** developing automated validation, benchmarking, profiling, and performance-analysis solutions for distributed systems, AI/ML workloads, cloud platforms, and compute-intensive applications.
- Strong programming expertise in **Python, C++, Java, and Linux**, with hands-on experience developing benchmark harnesses, profiling utilities, performance-regression infrastructure, automated test frameworks, and data-analysis tooling.
- Hands-on experience with **Machine Learning and Large Language Models**, including transformer architectures, training and inference lifecycles, model execution optimization, PyTorch, JAX, GPU/accelerator-backed workloads, and end-to-end ML performance characterization.
- Experienced analyzing **compute throughput, memory bandwidth, latency, utilization, cache behavior, DRAM/HBM performance, PCIe/DMA transfers, compute pipelines, and interconnect bottlenecks** using hardware performance counters and profiling tools.
- Strong understanding of **computer architecture**, including cache and memory hierarchies, HBM, DRAM, compute pipelines, accelerator execution, PCIe, DMA, interconnect topologies, collective communication, and distributed workload scaling.
- Experienced applying **statistical methods, regression analysis, benchmarking, performance modeling, and data visualization** to compare configurations, identify performance regressions, quantify bottlenecks, and drive optimization decisions.
- Proven experience across the full software development lifecycle, including coding standards, code reviews, Git-based source control, build systems, CI/CD, testing, operations, performance dashboards, root-cause analysis, and cross-functional collaboration with software and architecture teams.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Systems:** Python, C++, Java, Bash, Linux, Multithreading, Distributed Systems, Performance Engineering, Automation Tooling
- **Machine Learning & LLMs:** PyTorch, JAX, Transformer Models, Large Language Models (LLMs), Training, Inference, Model Execution Optimization, Vision Models
- **Accelerator & Kernel Technologies:** CUDA, GPU Computing, AWS Neuron, Trainium/Inferentia Concepts, TPU, ML Kernels, Low-Level Kernels, Accelerator Performance Profiling
- **Computer Architecture:** CPU/GPU/AI Accelerators, Caches, DRAM, HBM, Memory Hierarchies, Compute Pipelines, PCIe, DMA, Interconnect Topologies, Memory Bandwidth
- **Performance & Profiling:** Hardware Performance Counters, Linux perf, Nsight Systems, Nsight Compute, Throughput Analysis, Latency Analysis, Bottleneck Identification, Performance Regression
- **Distributed ML & Scaling:** AllReduce, AllGather, Collective Communications, Multi-Device Training, Interconnect Latency, Scaling Efficiency, Resource Utilization, Distributed Workloads
- **Analytics & Development:** Statistical Analysis, Regression Analysis, Pandas, NumPy, Data Visualization, Performance Dashboards, Git, CI/CD, Docker, AWS, Testing, Code Reviews, Build Processes

EXPERIENCE

Test Automation Engineer *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Designed and developed Python and C++ based **performance benchmarking and automated validation frameworks** to measure throughput, latency, resource utilization, scalability, and regression behavior across compute-intensive and distributed application workloads.
- Profiled AI/ML workloads using **PyTorch, JAX, Linux performance counters, Nsight Systems, and Nsight Compute**, analyzing execution timelines, kernel utilization, memory traffic, synchronization overhead, and compute bottlenecks.
- Executed representative **LLM, transformer, and machine-learning training/inference workloads** across accelerator-backed environments and compared throughput, latency, batch-size scaling, utilization, and memory-efficiency characteristics across configurations.
- Analyzed hardware and system performance across **cache hierarchies, DRAM/HBM bandwidth, PCIe transfers, DMA activity, compute pipelines, and interconnect communication**, isolating bottlenecks impacting end-to-end workload performance.
- Built automated performance-regression infrastructure and dashboards using Python, Pandas, NumPy, statistical analysis, and visualization tools to track benchmark trends, detect deviations, and compare results against defined performance targets.
- Applied regression analysis and statistical techniques to distinguish normal benchmark variance from meaningful performance regressions, correlating changes with software versions, workload parameters, resource configurations, and system behavior.
- Validated distributed workload scaling using **AllReduce, AllGather, collective communications, and multi-device execution**, measuring communication overhead, interconnect latency, synchronization cost, and scaling efficiency.
- Collaborated with software, architecture, infrastructure, and performance teams to reproduce bottlenecks, perform root-cause analysis, recommend optimization opportunities, and determine readiness of performance-sensitive software releases.

Environment: Python, C++, PyTorch, JAX, LLMs, Transformer Models, Linux, Linux perf, Nsight Systems, Nsight Compute, Hardware Performance Counters, CUDA, AWS Neuron, GPU/AI Accelerators, HBM, DRAM, Cache Hierarchy, PCIe, DMA, AllReduce, AllGather, Pandas, NumPy, Statistical Analysis, Regression Analysis, Performance Dashboards, Git, Docker, AWS, CI/CD

SDET *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Developed Python-based performance and reliability test infrastructure for distributed healthcare and AI-enabled services, measuring application throughput, latency, scalability, concurrency, resource utilization, and system stability under production-like workloads.
- Built benchmark workloads to evaluate ML inference and data-processing pipelines, profiling CPU/GPU execution, memory consumption, service latency, batch processing, and end-to-end workload efficiency across cloud environments.
- Used hardware and software profiling techniques to investigate **compute utilization, memory bandwidth, cache behavior, I/O latency, network overhead, and synchronization bottlenecks**, producing actionable performance findings.
- Created automated regression suites and performance dashboards that compared current benchmark results against historical baselines and release targets, highlighting latency, throughput, and resource-utilization regressions.
- Applied statistical analysis, sampling, variance analysis, and regression techniques to interpret noisy performance measurements, identify statistically meaningful changes, and support data-driven optimization decisions.
- Validated distributed application scaling and parallel processing behavior, analyzing communication overhead, workload partitioning, resource contention, asynchronous execution, and infrastructure constraints.
- Performed detailed root-cause analysis using backend logs, traces, system metrics, profiling output, SQL data, and cloud telemetry to isolate software and infrastructure contributors to performance degradation.
- Partnered with developers, platform engineers, data teams, and architects to improve system scalability, observability, reliability, and performance while integrating automated benchmark execution into CI/CD release workflows.

Environment: Python, Java, PyTorch, Machine Learning, Linux, Performance Profiling, Hardware Counters, CPU/GPU Workloads, Memory Bandwidth, Cache Analysis, Distributed Systems, AWS, Docker, CI/CD, Pandas, NumPy, Statistical Analysis, Regression Analysis, Performance Dashboards, Splunk, Git, Jenkins

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed Python, Java, and C++ utilities for automated system validation, workload benchmarking, and performance analysis across enterprise applications, backend services, and distributed computing environments.
- Designed performance test scenarios to measure application throughput, response latency, concurrency, scalability, CPU utilization, memory consumption, and I/O behavior under varying workload conditions.
- Analyzed low-level system performance using Linux profiling utilities and application instrumentation, identifying inefficient processing paths, excessive memory consumption, I/O constraints, and synchronization bottlenecks.
- Developed data-analysis scripts using Python, Pandas, NumPy, and SQL to aggregate benchmark results, identify performance trends, compare software configurations, and quantify regressions across release cycles.
- Applied statistical analysis and regression techniques to benchmark data, distinguishing workload variability from meaningful system-performance changes and providing evidence for optimization recommendations.
- Built automated CI/CD performance checks using Git, Jenkins, Maven, and reusable benchmark utilities to detect regressions before software promotion and maintain consistent performance baselines.
- Investigated performance defects using application logs, system metrics, traces, database records, and profiling data while collaborating with Software Engineers to isolate root causes and validate improvements.
- Participated in code reviews, design discussions, testing, release validation, and production support, contributing across the software development lifecycle and improving performance, reliability, and scalability.

Environment: Python, C++, Java, Linux, Performance Benchmarking, System Profiling, CPU/Memory Analysis, Pandas, NumPy, SQL, Statistical Analysis, Regression Analysis, Git, Jenkins, Maven, CI/CD, Distributed Systems, Root Cause Analysis, Performance Testing

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed Java and Python-based automated test and performance-validation utilities for enterprise web, mobile, API, and backend applications across distributed system environments.
- Executed performance, scalability, integration, system, regression, and end-to-end testing to measure application response times, throughput, concurrency behavior, resource consumption, and service reliability.
- Created workload-generation and automation scripts that simulated concurrent users, API traffic, backend processing, and high-volume application scenarios to identify performance degradation and capacity constraints.
- Analyzed CPU utilization, memory consumption, service latency, database response times, network behavior, application logs, and system metrics to identify performance bottlenecks and support root-cause analysis.
- Developed SQL and Python utilities to collect, process, and compare performance-test results across releases, supporting trend analysis, regression identification, and quality reporting.
- Integrated automated testing and performance-validation workflows with Git, Jenkins, Maven, and CI/CD pipelines to provide repeatable regression coverage and early detection of system-quality issues.
- Collaborated with Software Engineers and infrastructure teams to reproduce performance defects, evaluate proposed fixes, verify optimization improvements, and improve application reliability and scalability.
- Participated throughout the software development lifecycle, including requirements analysis, test design, code and test reviews, automation development, defect triage, regression testing, release validation, and production support.

Environment: Java, Python, Linux, Selenium WebDriver, REST APIs, Performance Testing, Scalability Testing, Distributed Systems, CPU/Memory Profiling, SQL, Jenkins, Git, Maven, CI/CD, Statistical Analysis, Regression Testing, Root Cause Analysis, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.